

Tyler Vander Mooren

Professor Mirza Mohd Shahriar Maswood

CYB 412 - Secure Hardware

April 2, 2026

Homework 5

Question 1

1.1 Explain the Classification of Physical Attacks and provide examples for each category

Physical attacks are commonly categorized into noninvasive, semi-invasive and invasive attacks, depending on how much physical access the attacker has to the hardware and how much they interfere with the device. Noninvasive attacks do not require opening the chip or damaging the device and instead this attack observes leakage from normal operation, such as timing attacks, power analysis and electromagnetic analysis. Semi-invasive attacks are different in that they usually require the chip package to be opened so the die is exposed, but they do not directly probe internal wires, with an example being laser fault injection. Invasive attacks are the most direct and destructive because they involve physically opening the device and probing, modifying or reconstructing the internal circuitry through methods such as microprobing, layer removal and reverse engineering. These attacks can also be viewed as passive or active, where the passive attacks generally only observe information leakage and active attacks intentionally disturb the system and in general the attack becomes more powerful as the attacker gains more direct access to the hardware (Bhunja & Tehranipoor, 2019).

Question 2

2.1 Explain the significance of "readback" in FPGA devices, its vulnerabilities, and differences between Xilinx and Altera's approaches to readback capabilities for defensive use against tampering and attacks

Readback in FPGA devices is the ability to read configuration data or internal state information back from the FPGA after it has been programmed. The significance in this is that it can support debugging, verification and defensive integrity checking against faults or tampering. The problem is that readback can also become a security vulnerability, for example if an attacker gains access to it then they may be able to recover configuration information, study the hardware design and support reverse engineering or cloning. Xilinx traditionally provides richer readback capability, which made it more useful for defensive checking and debugging, but it also meant that more internal information needed to be protected. Altera generally took a more limited or restricted approach in many of its devices, which did reduce the exposure of configuration data but also reduced some defensive flexibility. (Bhunja & Tehranipoor, 2019).

2.2 How do clock glitching and voltage glitching techniques work in fault injection attacks, and what are the potential consequences and objectives associated with these techniques in terms of compromising a target system's security and functionality?

Clock glitching and voltage glitching are both active fault injection attacks that intentionally try to disturb the normal operation of a device so it produces mistakes that help the attacker. Clock glitching works by disturbing the clock signal during a critical operation, which can introduce timing violations, skipped instructions, incorrect branching or unintended system states. Voltage glitching works by briefly lowering or spiking the supply voltage during an important computation. This can cause faulty reads, corrupted writes, wrong calculations or bit flips because digital circuits rely on stable voltage to operate correctly. The objective of both

attacks is usually not random damage but controlled malfunction, such as bypassing authentication, skipping secure boot checks or creating faulty cryptographic outputs that may reveal secret keys. The consequences can include memory corruption, crashes, skipped security checks and an overall loss of trust in the hardware system (Bhunia & Tehranipoor, 2019).

Question 3

3.1 Explain FPGA and the need of FPGAs. Also mention the applications of FPGA'

An FPGA, or Field-Programmable Gate Array is an integrated circuit that can be programmed by the user after manufacturing so it can implement different digital circuits depending on the application instead of being permanently fixed for one task. This is needed because it provides a balance between flexibility and performance, giving much better hardware level speed than software for many tasks while still being reprogrammable unlike ASICs. This reason alone makes them valuable when a design requires updates, testing or later changes without the cost and permanence of building a custom chip. FPGAs are utilized in embedded systems, telecommunications, digital signal processing, cryptographic hardware, aerospace systems, automotive electronics, prototyping, hardware acceleration and security research because they allow designers to test and deploy custom hardware designs with much more flexibility than fixed-function devices (Bhunia & Tehranipoor, 2019).

3.2 Explain in detail about Reverse Engineering

Reverse engineering itself is the process of analyzing a hardware system in order to understand how it was designed and how it works. In hardware security it is used to recover the internal structure, behavior or design details of a chip, board or complete devices. This is done by studying interfaces and behavior at the system level, examining PCB traces at the board level or removing the package, stripping away layers and inspecting the silicon with specialized tools

at the chip level. The ultimate goal is to steal intellectual property, clone a product, discover vulnerabilities or prepare for later attacks such as tampering or key extraction. This is why reverse engineering poses such a serious threat in secure hardware. To defend against it, designers may use obfuscation, logic locking, bitstream encryption, tamper resistance and restricted debug access. But overall reverse engineering remains dangerous because it turns a finished product back into design knowledge that the manufacturer was trying to protect (Bhunia & Tehranipoor, 2019).

Question 4

4.1 Explain the bootstrapping process for a Controlled Physical Unclonable Function (CPUF) in generating challenge-response pairs.

A controlled Physical Unclonable Function, or CPUF is a PUF that includes extra control logic so the raw PUF cannot be directly accessed. This is important because unrestricted access cannot be allowed, since it could allow an attacker to collect many challenge-response pairs and then try to model the device. During the bootstrapping process, the CPUF is enrolled in a trusted environment where a challenge or input is applied and the PUF generates a response based on the unique physical manufacturing variations of that chip. Because of how these variations differ from device to device, the resulting response is unique to that specific device. The challenge-response pair is then recorded or verified during enrollment so it can later be used for authentication, with the main purpose being to securely establish valid challenge-response behavior without exposing the raw PUF too openly (Gassend et al., 2008).

4.2 Explain the manufacturer's role in providing inputs, the CPUF's response generation, and the utilization of computed challenges to secure the (Challenge, Response) pair

The manufacturer plays a trusted role during the CPUF enrollment by providing the initial input or challenge information when the device is first set up and this input is then used through the control logic to manage the actual challenge that reaches the PUF core. The CPUF then generates a response based on the unique physical characteristics created by the manufacturing variation, which makes the response difficult to copy or predict exactly. A major feature of the CPUF is that it does not simply just expose raw challenge-response behavior, but instead it computes or processes challenges that are used through the control logic so that the true PUF interaction is protected. This helps secure the challenge-response pair by limiting the direct access to the raw PUF and reducing the overall risk of modeling or replay attacks, allowing the manufacturer to establish trusted challenge-response information that can then be used later securely for authentication (Gassend et al., 2008; Herder et al., 2014).

Question 5

5.1 What distinguishes Ring Oscillator (RO) Physical Unclonable Functions (PUFs) from Arbiter PUFs, emphasizing their advantages?

Both RO PUFs and Arbiter PUFs are delay based PUF designs, but they operate in different ways. An Arbiter PUF sends the signals through two delay paths and then uses an arbiter to determine which signal arrives first, while an RO PUF uses multiple ring oscillators and then compares their oscillation frequencies. In both cases, the response depends on tiny manufacturing variations, however RO PUFs do have a major practical advantage in that they are generally easier to implement on FPGA platforms. While Arbiter PUFs are typically harder to implement reliably on FPGAs because routing differences can interfere with the delay symmetry they depend on. So, RO PUFs are often considered more practical for reconfigurable hardware,

where the Arbiter PUFs are usually better suited for ASIC style designs where those paths can be controlled more precisely (Bhunias & Tehranipoor, 2019; Herder et al., 2014).

5.2 Explain the bit generation capability of RO PUFs and their susceptibility to reliability issues due to environmental factors. Discuss strategies to enhance RO PUF reliability and specify their suitability for different platforms compared to Arbiter PUFs

RO PUFs generate bits by comparing the frequencies of two ring oscillators, where one oscillator running faster than the other may produce a 1 and the opposite comparison may produce a 0. By comparing the many oscillator pairs the design then can generate multiple response bits. The main weakness is reliability because oscillator frequencies can be affected by temperature, supply voltage and aging. So, if two oscillators are very close in frequency the output bit may flip under different environmental conditions. Improvements that can be made to reliability would be to choose oscillator pairs with larger frequency differences, using repeated measurements, majority voting, error correction and carefully layout placement so the responses remain more stable. In short, because of the implementation characteristics of RO PUF, it is usually more suitable for FPGA platforms, while Arbiter PUFs are generally more suitable for ASICs where delay paths can be controlled more accurately (Bhunias & Tehranipoor, 2019; Herder et al., 2014).

References

Bhunias, S., & Tehranipoor, M. H. (2019). *Hardware security: A hands-on learning approach*.

Morgan Kaufmann.

Gassend, B., Dijk, M. V., Clarke, D., Torlak, E., Devadas, S., & Tuyls, P. (2008). Controlled physical random functions and applications. *ACM Transactions on Information and System Security*, 10(4), 1–22. [10.1145/1284680.1284683](https://doi.org/10.1145/1284680.1284683)

C. Herder, M. -D. Yu, F. Koushanfar and S. Devadas, "Physical Unclonable Functions and Applications: A Tutorial," in *Proceedings of the IEEE*, vol. 102, no. 8, pp. 1126-1141, Aug. 2014, [10.1109/JPROC.2014.2320516](https://doi.org/10.1109/JPROC.2014.2320516)